4bit $x_3x_2x_1x_0$ % 4 = 0

TV

$$z = x_0 \overset{\text{OR}}{+} \overset{\text{AND}}{\bar{x}_1\bar{x}_0}$$

KARNAUGH

$$z = \bar{x}_1 + x_0$$

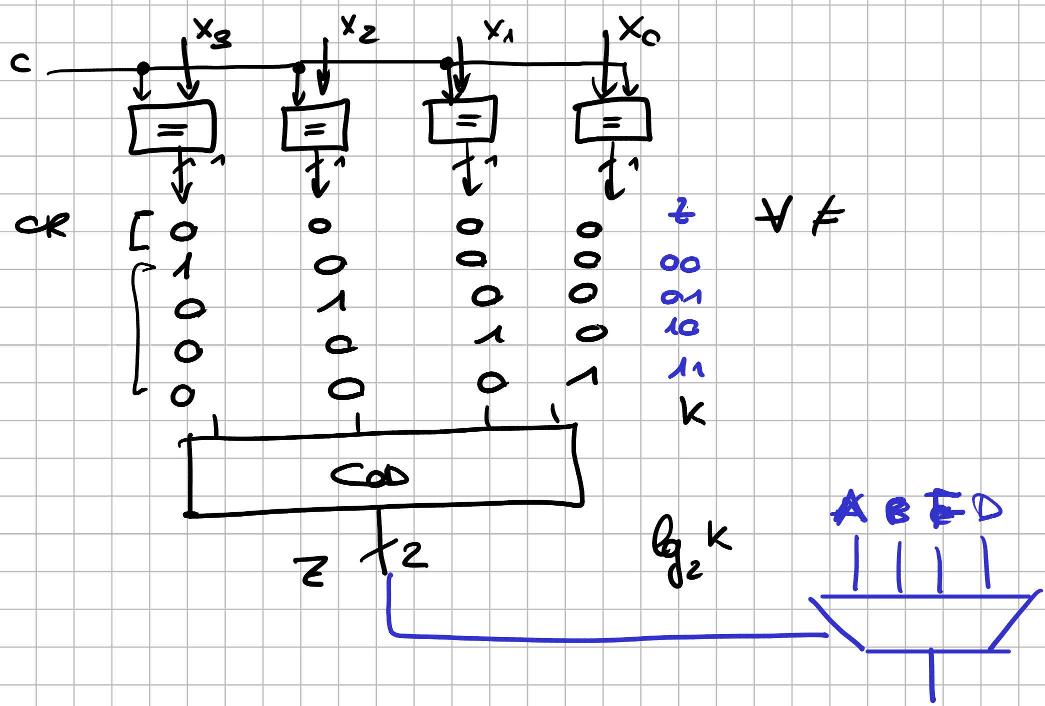
distrib $\Downarrow$

$$(x_0 + \bar{x}_1)\cdot(x_0 + \bar{x}_0)$$

$$\cancel{x + \bar{x}_0}$$

{ FA, MUX, DEMUX, CONFRONTATORE (=), CODIFICATORE (MUL), SHIFTER }

CODIFICATORE



TV COD

x_3	x_2	x_1	x_0	z_1	z_0
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1

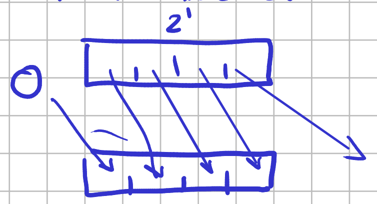
$$z_1 = \bar{x}_3\bar{x}_2 x_1\bar{x}_0 + \bar{x}_3\bar{x}_2\bar{x}_1 x_0$$

$$z_2 = \bar{x}_3 x_2\bar{x}_1\bar{x}_0 + \bar{x}_3\bar{x}_2\bar{x}_1 x_0$$

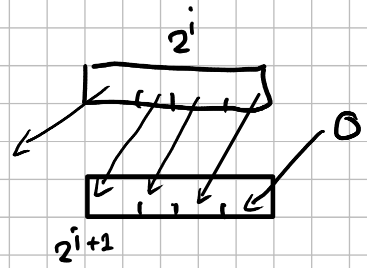
SHIFTER

SHIFT { logici / aritmetici }   { → ds / ← sn }

SHIFT LOGICI



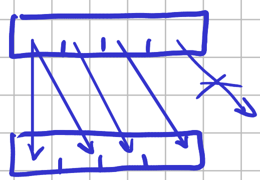
shift logici ds di 1 pos   /2



shift logico a sn di 1 pos   *2

~~sn~~

SHIFT ARITMETICI : ds

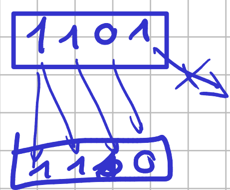


-3

$2^{4-1} \,..\, 2^{4-1}-1$

$|-3|$ 0011

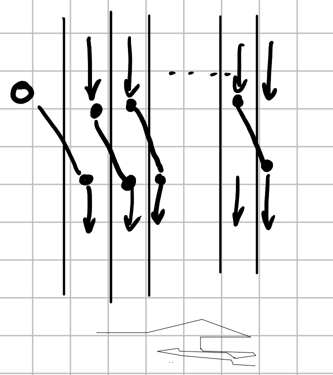
1100



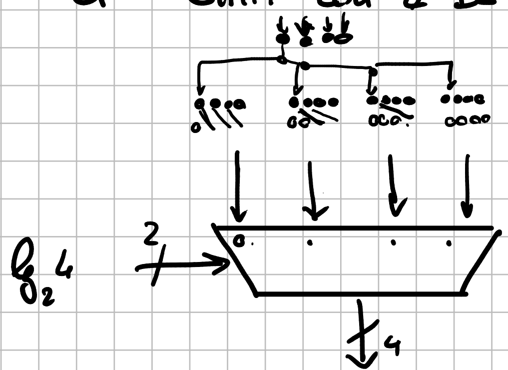
1110
0001
$-$ 1
0010

1100
0001
1
0010
$-$

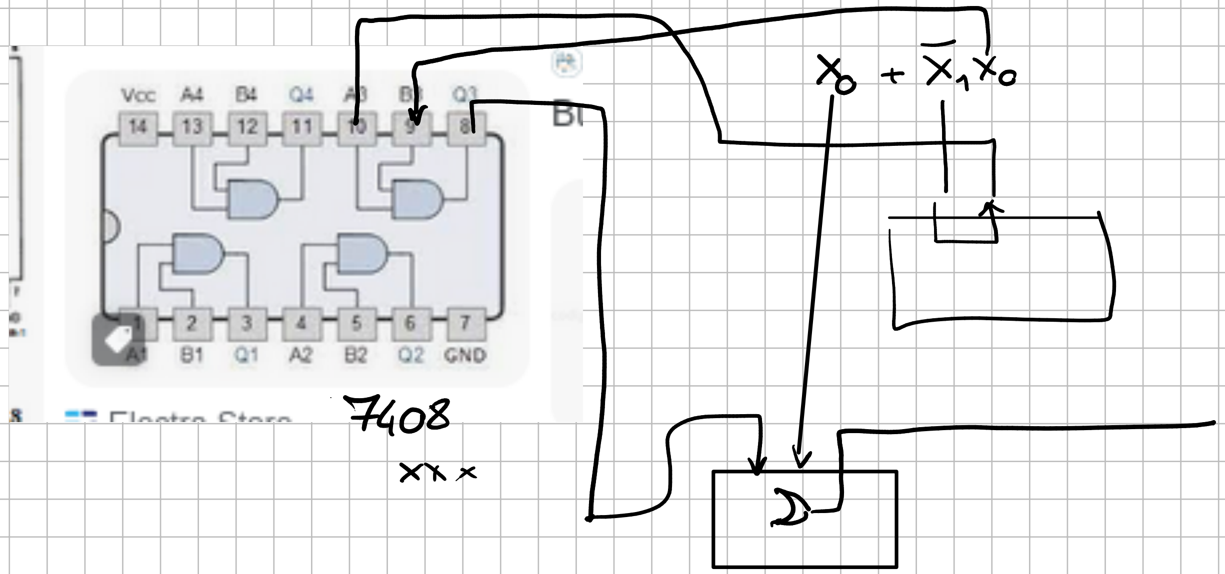
SHIFT LOG DS 1

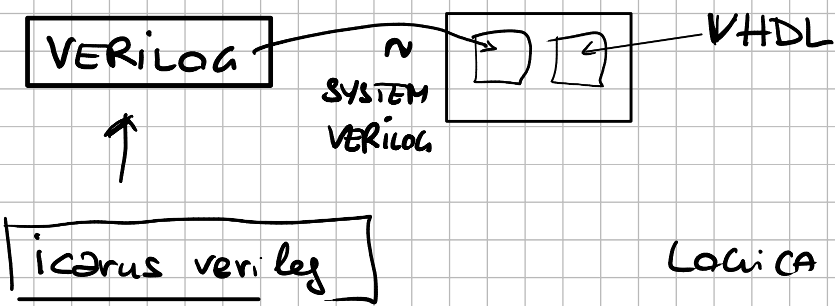


OP SHIFT LOG a DS di K pos



k=4





LOGICA COMBINATORIA

FUNZIONI BOOLEANE

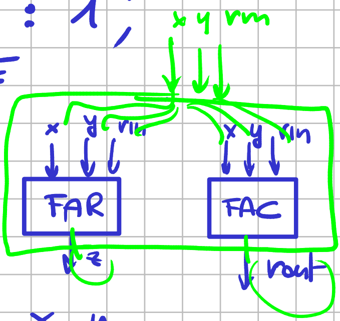
1 solo bit output

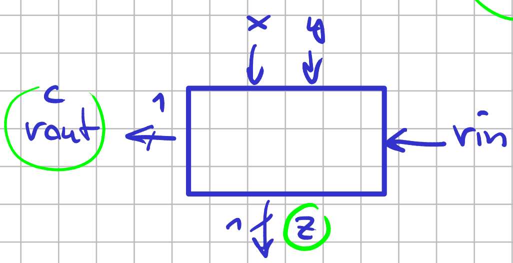
PRIMITIVE name ( );

tabella verità

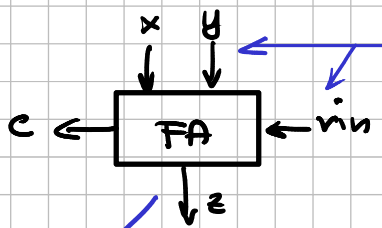
ENDPRIMITIVE

TABLE
0 0 0 : 0;
0 0 1 : 1;
⋮
1 1 1 : 1;
ENDTABLE





MODULE name ( );



input x, input y,
input rin,

output z,
output c

2^k

input [0:1] k → $k_0 k_1$

[1:0] k → $k_1 k_0$

[9:10] k → k[9] k[10]

ENDMODULE

assign varout = $\underbrace{\text{expr bool}}_{\text{delle var di input}}$

&& AND

! NOT

|| OR

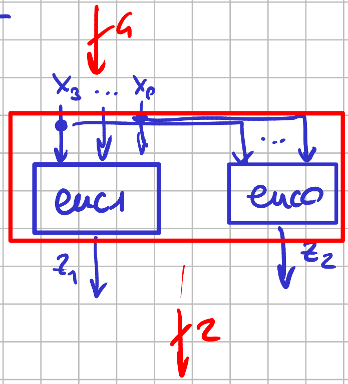
assign z = (x&&y)||(!x &&
!y);

TV COD

$x_3 x_2 x_1 x_0$	$z_1 z_0$
1 0 0 0	0 0
0 1 0 0	0 1
0 0 1 0	1 0
0 0 0 1	1 1

4 ingressi da 1 bit

$\}$ 2^4 righe



$$z_1 = \bar{x}_3 \bar{x}_2 x_1 \bar{x}_0 + \bar{x}_3 \bar{x}_2 \bar{x}_1 x_0$$

$$z_2 = \bar{x}_3 x_2 \bar{x}_1 \bar{x}_0 + \bar{x}_3 \overline{x_2 x_1} x_0$$

$x_1 x_0 \overset{?}{\geq} y_1 y_0$

f: qual'è il più grande di 2 numeri a 2 bit

$x_1 x_0$	$y_1 y_0$	$z_1 z_0$
0 0	0 0	0 0
	0 1	0 1
	1 0	1 0
	1 1	1 1
0 1	0 0	0 1
	0 1	0 1
	1 0	1 0
	1 1	1 1
1 0		
1 1		

z_1

$x_1x_0 \backslash y_1y_0$	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	1	1	1	1
10	1	1	1	1

z_0

$x_1x_0 \backslash y_1y_0$	00	01	11	10
00	0	1	1	0
01	1	1	1	0
11	1	1	1	1
10	0	0	1	0

$$z_1 = x_1 + y_1$$

$$z_2 = x_1 x_0 + y_1 y_0 + \bar{x}_0 y_0 + x_0 \bar{y}_1$$

module maggiore di (output [1:0] z,
input [1:0] x, input [1:0] y);